Predicting the Zero-Shot Transfer of a Promptable Video Tracker

Forecasting SAM 3's per-species, per-place reliability —
for detection and association — from label-free distances

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Abstract

Promptable trackers such as SAM 3 segment and track any named animal in a video with no training on that species, yet no rule tells a practitioner *in advance* whether the model is reliable for a given animal in a given place. This dissertation builds and validates a *predictive* rule for that reliability on the SA-FARI benchmark, whose splits are disjoint by species and location. We measure SAM 3's zero-shot promptable-tracking accuracy with the official evaluator, decompose it into detection (`pDetA`) and association (`pAssA`), and test whether each can be predicted from four label-free distances (taxonomic, visual, environment, temporal). *[TODO: headline result + validated out-of-sample performance.]*

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Contents

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Motivation

Promptable video trackers such as SAM 3 [4] segment and track any named animal in a video with no training on that species, and conservation teams now deploy them on new species and regions constantly. Yet no rule tells a practitioner, *before running the model*, whether its output is trustworthy for a given animal in a given place.

1.2 Problem statement and research question

This dissertation asks whether SAM 3's zero-shot tracking accuracy on an unseen (species, place) can be *predicted in advance* from properties measurable without any label of the target, and which factors govern that transfer — separately for *finding* the animal and *following* it.

1.3 Hypotheses

H0 (null)
: the label-free distances carry no predictive information about `pDetA`/`pAssA` — transfer is not distance-driven.

H1 (detection)
: `pDetA` falls with species novelty (taxonomic + visual distance).

H2 (association)
: `pAssA` falls with environment difficulty (scene, day/night/IR, clutter, camera motion).

Failing to reject H0 would itself be an informative result, motivating a pivot from distance-based prediction to representational probing (does SAM 3 encode taxonomy?).

1.4 Contributions

To our knowledge this is the first study to *predict*, before the model is run and from label-free distances, the zero-shot transfer of a promptable video tracker — and the first to split that prediction into **detection** (`pDetA`) and **association** (`pAssA`). This distinguishes it from label-free performance prediction for classifiers, which estimates a single scalar from the model's own outputs on the target [1, 6, 10], and from the very recent static-image detector/segmenter estimators [14, 16]; and from camera-trap shift studies, which measure but do not forecast per-site reliability [2, 3, 7]. Concretely we deliver: (i) a validated, out-of-sample reliability estimator for SAM 3 over SA-FARI; (ii) the detection-versus-association decomposition, tested on two near-orthogonal splits — a constructed species hold-out (novelty) and the official location hold-out (environment); and (iii) an honest accounting of when distance *fails* to predict transfer (H0).

1.5 Dissertation outline

Chapter 2 reviews the related work; Chapter 3 describes the data, measurement, distance features, and modelling; Chapter 4 reports the results and out-of-sample validation; Chapter 5 discusses them; Chapter 6 concludes.

Chapter 2

Background and Related Work

This chapter places the dissertation among four bodies of work: promptable segmentation and tracking (the model we study), the tracking metric and its detection/association decomposition (how we score it), label-free prediction of out-of-distribution performance (the statistical idea we extend), and distribution shift in camera-trap vision (the application and the closest empirical neighbours). It closes by stating precisely the gap this work fills.

2.1 Promptable segmentation and tracking

The *Segment Anything* line moved segmentation from a fixed label set to an interactive, promptable formulation, and its successors extended prompting from points and boxes to *concepts*. SAM 3 [4] is a promptable model that, given a short text prompt such as `"impala"`, detects, segments, and tracks every matching instance across a video with no training on that category — a *zero-shot*, open-vocabulary detect–segment–track capability. It is exactly this capability that conservation teams now deploy on new species and regions, and it is the (frozen) model whose transfer we predict. SAM 3 was released alongside SA-FARI [15], a large multi-animal camera-trap tracking dataset that we use as the substrate for our natural experiment (Section 2.6). We treat SAM 3 strictly as a fixed black box: we never fine-tune it, and the only quantity we fit is a small regression *about* its behaviour.

2.2 The tracking metric and its decomposition

Higher Order Tracking Accuracy (HOTA) [9] evaluates multi-object tracking by combining, over a range of localisation thresholds, a *detection* term (`DetA`: were the right objects found?) and an *association* term (`AssA`: were identities kept consistent over time?). SA-FARI scores promptable tracking with a promptable variant, `pHOTA`, that decomposes identically into `pDetA` and `pAssA`. This decomposition is the intellectual core of the dissertation: rather than predict a single scalar, we ask whether *finding* and *following* an animal transfer for *different reasons* — the former with how novel the animal is, the latter with how difficult the scene is — and we therefore model the two components separately throughout.

2.3 Predictable out-of-distribution generalisation

A now-mature literature shows that a model's performance under distribution shift can be anticipated *without* target labels. Miller et al. [10] observe that out-of-distribution (OOD) accuracy is strongly linearly correlated with in-distribution accuracy across many models and shifts, and Baek et al. [1] show that the *agreement* between a pair of models tracks OOD accuracy closely enough to estimate it from unlabelled data alone. A parallel strand estimates accuracy from a model's own statistics on the unlabelled target: average thresholded confidence [6] calibrates a confidence threshold on source data and reads off target accuracy as the mass above it, while Deng and Zheng [5] regress classifier accuracy directly on a *distribution-shift feature* — the Fréchet distance between train and test feature sets — which is the closest methodological cousin to our own "regress the score on a distance" approach.

Two properties characterise essentially all of this work, and both are limitations we depart from. First, it targets **image classifiers** with a single scalar accuracy. The idea has only recently reached structured outputs: Yoo et al. [16] estimate object-detection performance without ground truth from the spatial consistency and confidence reliability of pre-NMS boxes, and reverse classification accuracy [14] predicts per-image segmentation quality by training a reverse model against a labelled reference. These are the closest structured-output predecessors, but both operate on *static single images*, neither concerns identity over time, and none decomposes the estimate into detection versus association. Second, every one of these methods still *runs the model on the target* and inspects its predictions, confidences, or agreement — it estimates the score of an *already-executed* model. We instead forecast from properties knowable *before the model is ever run* on the target.

2.4 Distribution shift in camera-trap and wildlife vision

That camera-trap models degrade at new locations is well established. *Terra Incognita* [2] showed that recognisers excellent at trained camera sites fail sharply at unseen ones; WILDS [8] consolidated this (via iWildCam) as a canonical real-world shift; and PanAf-FGBG [3] demonstrated that behaviour-correlated *backgrounds* causally drive the drop, proposing a latent-space mitigation. These works are the empirical backdrop for our environment axis — but they share two features that mark the gap. They **measure or mitigate** degradation after the fact rather than *predicting* per-site or per-species reliability in advance, and they study **classifiers or behaviour recognisers**, not promptable trackers, scored by a single accuracy with no detection/association split. Most tellingly, a recent unified study of camera-trap recognition over time [7] explicitly names "predicting when zero-shot models will succeed at a new site" as an open question — and stops at naming it. This dissertation builds the predictor that question asks for.

2.5 What should predict transfer

Our candidate predictors are motivated by representation-learning results. Vision foundation models encode taxonomic structure across the tree of life [13], which motivates a *taxonomic* distance and, because appearance and phylogeny are correlated, a *visual* distance; we compute the latter from mask-cropped DINOv2 [11] embeddings, with CLIP [12] as a robustness variant. The camera-trap location results above [2, 3] motivate an *environment* distance from the scene background, and per-video timestamps motivate a *temporal* gap. Crucially, all four are computable without any label of the target animal, and all are measured against a frozen reference set, so no target information can leak into a predictor.

2.6 The SA-FARI benchmark

SA-FARI [15] is a large open multi-animal camera-trap tracking dataset (thousands of densely annotated videos across roughly 80 annotated species with a seven-level taxonomy, hundreds of locations on four continents, and per-video timestamps), released with SAM 3 and scored by the official `VEval` evaluator. It ships with abundant *hard negatives* — prompts for absent species that must return nothing — which we keep, as they drive the false-positive side of detection. Two properties of the shipped split shape our design and merit a precise statement. Inspection of the released annotations shows the train/test split is disjoint by **location** (test camera sites are unseen) but that its *species are largely shared* across the split; a genuine “unseen species” axis is therefore not directly available from the official split. Because SAM 3 is frozen and zero-shot on *all* of SA-FARI, however, the split reflects nothing the model learned — it is only our choice of a *reference distribution* against which distances are measured. This licenses a species-disjoint *reference/probe partition* of the same videos for the primary (species-novelty) experiment, complemented by the official location split for the environment experiment; the two are near-orthogonal probes of the same data, and their construction is detailed in Chapter 3.

2.7 Summary and gap

Predictable OOD generalisation is established for classifiers scored by a scalar [1, 5, 6, 10] and, very recently, for static detectors and segmenters [14, 16]; camera-trap vision has thoroughly documented, but not forecast, location-driven degradation [2, 3, 7, 8]. No prior work predicts, *before running the model* and from label-free distances, the zero-shot transfer of a **promptable video tracker**, and none separates that prediction into **detection** and **association**. Filling exactly that gap — a validated, out-of-sample reliability estimator for SAM 3 over SA-FARI's species-by-place-by-time structure, decomposed into `pDetA` and `pAssA` — is the contribution of this dissertation.

Chapter 3

Data and Methodology

3.1 Dataset, splits, and the analysis unit

We use SA-FARI [15] throughout. In the released `_ext` annotations, a probe is a (video, text-prompt) pair; the animal's species and its seven-level taxonomy are resolved through the prompt's `category_id`, while per-video `location_id` and creation timestamp give place and time. We aggregate scores to a *cell* — a (species, location, time) group — and attach support counts (n_{frames}, n_{masklets}, n_{videos}) used for weighting and as covariates. Hard negatives (prompts for absent species) are kept, as they drive the false-positive side of detection.

Two complementary splits. Inspection of the annotations shows the shipped split is disjoint by *location* but shares species across train and test (Section 2.6); on that split species-distance is ≈ 0 for every cell, so it cannot test species novelty. Because SAM 3 is frozen and zero-shot on all of SA-FARI, the split only defines our *reference distribution*, so we are free to repartition the same videos. We therefore use two near-orthogonal experiments and run inference only once (scoring is per-video and split-agnostic):

Split A — species hold-out (primary).
The pooled species are partitioned into a reference set and a held-out set, stratified by taxonomic group so held-out species span a range of novelty. Species-distance varies while the environment stays largely familiar (held-out species recur at reference locations), isolating H1. Distances use a *leave-species-out* reference and it is validated by leave-species-out cross-validation.

Split B — location hold-out (secondary).
The official train/test split, whose unseen locations isolate H2 and whose benchmark comparability carries the Phase-1 score-sanity check.

Split A is a *constructed* hold-out — legitimate because the model is frozen, and reported as such; we cross-check aggregate scores against Split B, which is benchmark-comparable.

3.2 Measuring transfer

SAM 3 [4] is run frozen over the probe videos (species-specific prompts as the primary condition, a generic "animal" prompt for robustness) and scored with the official `VEval` evaluator — the metric is never re-implemented — giving per-cell `pDetA`/`pAssA`/`pHOTA` [9] with their support. A single inference pass over the union of Split A and Split B probes serves both experiments.

3.3 The frozen reference (leakage firewall)

For each split the reference ("seen") species, locations, and taxonomy are frozen once; every distance is then computed against that reference only, so no probe information can enter a feature — verified by a unit test that swaps the reference and checks the distances move as expected. We assert the axis that is genuinely disjoint (location for Split B; the held-out species for Split A) and *report* any residual species overlap honestly rather than asserting it away. On Split A, distances use a leave-species-out view of the reference, so a held-out species' distance is to the nearest *other* reference species.

3.4 Label-free distance features

All four distances (plus a proxy) are label-free for the target and computed against the frozen reference.

3.4.1 Taxonomic distance

Lowest-common-ancestor tree steps from the probe's species to the nearest reference species (shared genus = 1, family = 2, order = 3, ...); the leave-species-out reference keeps this non-trivial for shared species.

3.4.2 Visual distance

Cosine distance between DINOv2 [11] embeddings of the *ground-truth-mask-cropped* animal and the nearest reference-species prototype, with a Fréchet/MMD variant and a CLIP [12] encoder as robustness checks. Using the dataset's ground-truth masks means this feature needs no SAM 3.

3.4.3 Environment distance

Embedding distance between the animal-masked-out background scene and the nearest reference location, plus an interpretable day/night–infrared flag derived from colour statistics.

3.4.4 Temporal gap

The minimum absolute year gap between a cell's footage and the nearest reference footage.

3.4.5 SAM 3 familiarity proxy

Separability of the target species in SAM 3's own feature space — a hedge against the unknown true pretraining set, reported alongside the other distances rather than treated as ground truth.

3.5 Modelling the law

Each bounded target (`pDetA`, `pAssA`) is modelled *separately* with a support-weighted beta / logit-link regression that includes a $\log(n_{\text{frames}})$ covariate, so that "rare" is never mistaken for "far". Variance is attributed to each factor by dominance/commonality analysis with a VIF collinearity check — not by four univariate fits — and the headline contrast (does detection load on species novelty and association on environment?) is quantified across the two targets. The fitted models are validated *out of sample* by holding out whole groups: leave-species-out on Split A and leave-location-out on Split B, with bootstrap confidence intervals reported wherever a point estimate appears. The validated models are wrapped into the reliability estimator: four distances in, predicted `pDetA`/`pAssA` with an interval out.

3.6 Reproducibility and constraints

The model is frozen and scored by the official evaluator; the seen set is frozen before any distance is computed; hard negatives are retained; and all inference is single-GPU and inference-only. Every reported number is regenerated from the repository, and the split definitions, seeds, and reference sets are persisted so both experiments are reproducible end to end.

Chapter 4

Results and Evaluation

4.1 Measurement

Zero-shot SAM 3 on the SA-FARI test split reproduces the published ballpark (Gate 1); the support-weighted per-cell aggregate is in Table 4.1.

4.2 The fitted law

Table 4.2 gives the standardised logit-GLM coefficients for detection and association.

4.3 Variance partitioning and the decomposition test

4.4 Out-of-sample validation

Table 4.3 reports whole-group hold-out error against a mean-predictor baseline.

4.5 Ablations and robustness

Table 4.1: Zero-shot SAM 3 on the SA-FARI test split — support-weighted mean over 346 scored cells (of 1447).

Metric (mask)	Value
pDetA	0.527
pAssA	0.546
pHOTA	0.533

Table 4.2: Standardised logit-GLM coefficients with *group-cluster-bootstrap* 95% CIs (resampling whole species / locations, so the interval respects pseudo-replication and the support weighting), for detection (pDetA) and association (pAssA). Every distance's CI spans zero.

Factor	pDetA coef. [CI]	pAssA coef. [CI]
Taxonomic distance	-0.362 [-3.87, 0.09]	-0.368 [-3.40, 0.08]
Visual distance	-0.144 [-0.50, 0.14]	-0.122 [-0.49, 0.18]
Environment distance	0.042 [-0.19, 0.24]	0.013 [-0.23, 0.21]
Clutter	-0.066 [-0.38, 0.29]	-0.087 [-0.41, 0.28]
Night / IR	-0.405 [-0.94, 0.11]	-0.462 [-1.02, 0.11]
log(support)	-0.216 [-0.51, 0.17]	-0.224 [-0.53, 0.18]

Table 4.3: Out-of-sample error: whole-group hold-out (leave-species-out / leave-location-out) vs a mean predictor. Δ is the MAE gain over baseline; its bracketed 95% CI and one-sided p come from a paired bootstrap resampling whole held-out groups. No hold-out's gain is significant.

Hold-out	Target	n	MAE	Baseline	Δ [95% CI]	p
location	pAssA	346	0.289	0.295	+0.007 [-0.005, +0.018]	0.13
location	pDetA	346	0.293	0.298	+0.004 [-0.007, +0.015]	0.23
species	pAssA	346	0.289	0.295	+0.006 [-0.006, +0.017]	0.16
species	pDetA	346	0.293	0.298	+0.004 [-0.006, +0.015]	0.24

Chapter 5

Discussion

5.1 Interpretation

5.2 A deployment reliability estimator

5.3 Limitations and threats to validity

5.4 Ethical considerations

Chapter 6

Conclusion

6.1 Summary of findings

6.2 Contributions

6.3 Future work

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Appendix A

Reproducibility

Appendix B

Supplementary Results